Towards Next Generation Risk Assessment of PFAS. Development and evaluation of a mechanistic human PBK model for PFOA.
Step 1

Chrysanthi Pachoulide[a,1,*], Joost Westerhout[b,2], Nynke I. Kramer[a,3]

[a] *Wageningen University and Research, Division of Toxicology, Stippeneng 4, Wageningen, 6708 WE*

[b] *RIVM, Department Name, Street Address, Utrecht, Postal Code*

Abstract

Per- and polyfluoroalkyl substances (PFAS) represent a substantial chemical class, comprising thousands of structurally diverse chemicals, which only have in common one fully fluorinated methyl or methylene carbon atom. PFAS are ubiquitous environmental pollutants of anthropogenic nature and therefore of potential risk to human health. Human toxicological risk assessment of PFAS is challenging, due to their sheer number, structural diversity and paucity of the relevant data required to identify and characterize them. A solution to this challenge lies in the application of read-across approach, which involves the extrapolation of information from a data-rich PFAS to make predictions about a similar data-poor one. One example of this approach could be the application of a Physiologically Based Kinetic (PBK) model from PFOA, to inform the development of PBK models for PFASs with analogous physicochemical properties. Despite the availability of several PBK models for PFOA, all of them rely on PFOA-specific animal-sourced data and are optimised to correctly predict the long half-lives observed in humans. This hinders their use in the read-across scope, where PBK models should be as mechanistic as possible, with key toxicokinetic processes are described based on either in vitro or in silico studies. The present study introduces a human mechanistic PBK model for PFOA, that uses only in vitro derived input parameters. The uptake and elimination of PFOA are predicted using in vitro to in vivo extrapolation principles, while its tissue

* Corresponding author

Email addresses: chrysanthi.pachouide@wur.nl (Chrysanthi Pachoulide), joost.westerhout@rivm.nl (Joost Westerhout), nynke.kramer@wur.nl (Nynke I. Kramer)

[1] This is the first author footnote.

[2] Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

[3] Yet another author footnote.

distribution is predicted using experimentally derived distribution coefficients to tissue components, such as albumin and phospholipids. The model introduces a minimal mechanistic kidney compartment to predict the active re-absorption of PFOA in the proximal tubules and a mechanistic transporter-mediated enterohepatic circulation. Furthermore, given the high affinity of PFOA to albumin, the unbound fraction determined in plasma, is adjusted to the differential albumin concentration between the plasma and tissue or tubular fluid compartments, to better describe an albumin facilitated transport of PFOA. The predicted Area Under the Curve (AUC) and half-life were compared to the concentration over time curve from a human volunteer study as well as to observed plasma concentrations and half-lives from the latest human biomonitoring data. Finally, the Global Sensitivity Analysis (GSA) workflow recommended by the OECD PBK guideline was applied to both evaluate the validity of the model and inform about the key parameters determining the PBK model predictions. The newly developed PFOA PBK model can be extrapolated to other PFASs for which these key parameters are already available. The subsequent derivation of these parameters for other PFASs will facilitate the extrapolation of the present PBK models to them as well and enable their human toxicological risk assessment.

Keywords: PFOA, Physiologically Based Kinetic modelling, read-across, Global Sensitivity Analysis, Next Generation Risk Assessment

---

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) represent a substantial chemical class, comprising thousands of structurally diverse chemicals, which only have in common one fully fluorinated methyl or methylene carbon atom. PFAS are ubiquitous environmental pollutants of anthropogenic nature and therefore of potential risk to human health. Human toxicological risk assessment of PFAS is challenging, due to their sheer number, structural diversity and paucity of the relevant data required to identify and characterize them. A solution to this challenge lies in the application of read-across approach, which involves the extrapolation of information from a data-rich PFAS to make predictions about a similar data-poor one. One example of this approach could be the application of a Physiologically Based Kinetic (PBK) model from PFOA, to inform the development of PBK models for PFASs with analogous physicochemical properties. Despite the availability of several PBK models for PFOA, all of them rely on PFOA-specific animal-sourced data and are optimised to correctly predict the long half-lives observed in humans. This hinders their use in the read-across scope, where PBK models should be as mechanistic as possible, with key toxicokinetic processes are described based on either in vitro or in silico studies. The present study introduces a human mechanistic PBK model for PFOA, that uses only in vitro derived input parameters.

Too many PFAS PFOA toxicokinetic understanding long half-life, persistence strong binding to proteins (especially albumin and fatty-acid binding proteins) and membrane phospholipids Discussion about previous PBK approaches, need of fitting to data & use of parameters taken from animal studies - read-across not possible - What we need: more generic, that we can use for PFAS for which we don't know information

2. Materials and Methods

2.1. PBK Model Development

The PFOA human PBK model was developed with the aim of being as minimal and mechanistic as possible for read-across in a NGRA framework. For this, the critical toxicokinetic processes relevant for PFAS were included, those being renal elimination and re-absorption, enterohepatic circulation and oral, dermal or inhalation exposure. The model was evaluated against observed *in vivo* concentration-time profiles in plasma as well as observed half-life in humans, from human biomonitoring data.[4]

2.1.1. Model structure

The PBK model was build on the model previously used by EFSA and recently updated to include dermal exposure (Husøy et al., 2023). The model was updated to include physiological processes related to renal elimination and re-absorption, as well as enterohepatic circulation. The model structure included the intestinal, kidney and liver compartments which are responsible for the above-mentioned toxicokinetic processes and the option to include skin and lungs in case relevant exposure routes are used.

2.1.1.1. Absorption.

2.1.1.1.1. Oral absorption

Oral absorption via food or drinks is the main route of exposure to PFOA. PFOA is quickly and efficiently absorbed from the gastro-intestinal tract to portal circulation (Abraham et al., 2024)[5]. A most physiological and mechanistic model for the gastro-intestinal tract (GIT) was initially considered to fully describe the toxicokinetic process involved in the absorption and elimination of PFOA in that organ. PFOA could be absorbed in all sections of the GIT both by passive diffusion of the fraction non-ionised and active uptake of the fraction ionised at the pH of the corresponding section. Uptake via both passive and active processes has been also observed in *in vitro* uptake studies (Kimura et al., 2017; Janssen et al., 2024) . The initial model subdivided the GIT into the stomach, upper small intestine (duodenum and jejunum), lower small intestine

[4] Could consider also evaluation against urine and fecal concentrations from Abraham study, but in this study they mentioned that urinary PFOA concentrations were contaminated while fecal concentrations (ng/g feces) don't represent fecal daily excretion and can't be used for data evaluation

[5] Plasma Tmax of 2h after oral exposure (from Abraham)

(ileum) and large intestine. Each intestinal sub-section was further divided into the lumen and intestinal vascular tissue. Absorption of PFOA was modelled as an uptake clearance from the lumen to the intestinal tissue, while the fraction of PFOA that was not absorbed in each section was transferred to the following section, or eliminated to the faeces, based on physiological transit times (Willmann et al., 2004). In the final model, the GIT was simplified to only one compartment as the initial more complex model did not significantly altered the absorption profile of PFOA[6].

2.1.1.1.2. Dermal absorption

Dermal absorption can occur after exposure via cosmetic products or dust disposed on skin. Dermal absorption was modelled using a compartmentalised skin compartment with a separate skin barrier (epidermis) and rest of skin compartment. Absorption of PFOA from the skin barrier was defined as the absorption clearance[7], while once PFOA was in the rest of skin compartment it was then distributed to the systemic plasma concentration.

2.1.1.1.3. [Inhalation absorption]Inhalation absorption[8]

Fast absorption, with Tmax at 3h post exposure (rats) (Gustafsson et al., 2022)

PFAS delivery on lung surfactants facilitated by albumin (Forsthuber et al., 2020)

For air partition coef (Yao et al., 2025)

2.1.2. Distribution

Tissue distribution of PFOA was assumed to be perfusion limited, as not enough mechanistic data is available to support a permeability limited model. Permeability limited PBK models for PFOA have been already developed, but they under-predict plasma concentrations and require the addition of a fitted parameter to correctly describe a reabsorption process from cells to plasma (Lin et al., 2023; Cheng and Ng, 2017). In previous PFOA PBK models, tissue-plasma partition coefficients were determined from *in vivo* animal steady-state tissue and plasma concentrations. This approach has been recently proven not adequate due to significant inter-species variabilities in the toxicokinetics of PFOA[9]. An alternative approach to defining tissue-plasma partition coefficients, is the mechanistic tissue-plasma partition algorithms, that use physiochemical

[6] Insert reference to predictions in supplementary material

[7] Is it correct to call it a clearance? I perceive clearance as an elimination parameter, but I could be wrong.

[8] tentative...

[9] refer to Deepika/Annelies study

properties (such as the octanol-water partition coefficient Kow and acid disociation constant pKa) to predict the chemical affinities tissue components (proteins or lipids), assuming that these depend on the lipophilicity and ionisation status of the chemical. The final steady-state tissue-plasma distribution is based on the differences in composition of the different tissues and their respective volume (Rodgers and Rowland, 2007; **?**; Peyret et al., 2010). However, these algorithms are limited to the chemical-space for which they have been validated for and cannot be applied to PFAS. PFAS, due to their amphiphilic nature lie outside of the applicability domain of these algorithms as they don't have a Kow. Recently, the binding affinity of PFOA to different tissue components (membrane phospholipids, storage lipids, structural proteins, albumin, fatty-acid binding protein) was derived experimentally (Allendorf et al., 2020). PFOA showed high affinity for albumin, fatty acid binding proteins and membrane phospholipids, while it's binding affinity to structural proteins and storage lipids was lower. Tissue-plasma partition coefficients were calculated using the above mentioned distribution coefficients. Physiological data about the relevant tissue components were taken from literature (Allendorf et al., 2020; Utsey et al., 2020). To enable comparison of the calculated partition coefficients with the previously used *in vivo* derived ones, partition coefficients were derived for both rat and human tissues. Results are available in supplementary file...

2.1.3. Elimination

Renal and biliary elimination of PFOA has been well documented, while renal and intestinal reabsorption are believed to be the mechanisms determining its long plasma half-life.

2.1.3.1. Renal elimination.

Multiple mechanisms can be involved in the renal elimination of PFOA, including glomerular filtration, active tubular secretion and re-absorption and passive permeability. Several studies have showed that PFOA is a substrate of organic anion transporters, while recently its intrinsic clearance by these transporters was determined in vitro, using either HEK293, or CHO cells transfected with the specific transporter (Yang et al., 2010; Janssen et al., 2024; Ryu et al., 2024; Lin et al., 2023). PFOA can be actively secreted from the plasma to the proximal tubule cells by OAT1 and OAT3 transporters and actively re-absorbed from the primary urine to the proximal tubule cells by OAT4 and URAT1 transporters (Yang et al., 2010; Janssen et al., 2024; Ryu et al., 2024; Lin et al., 2023). The available un-ionised PFOA can also freely diffuse between plasma, cells and the primary urine. A mechanistic kidney was developed, based on models previously developed for drugs undergoing renal reabsorption (Scotcher et al., 2016; Huang and Isoherranen, 2018; Pletz et al., 2021). Briefly, the proximal tubule was explicitly modelled as local PFOA concentrations are determining the extend of transporter mediated re-absorption. All other kidney sections (loop of Henle, distal tube and the collecting duct) were lumped together as rest-of-kidney as it was assumed that they don't significantly contribute to PFOA renal kinetics. Each section was initially sub-divided to plasma, cellular and luminar sub-sections where PFOA is distributed based on the active and or passive processes mentioned above. In the final model, the plasma and cellular compartments were lumped together to as active secretion by OAT1 and OAT3 transporters did not significantly contribute to the model predictions, while passive re-absorption from the proximal tubule cell to the plasma was not enough to accurately predict plasma concentrations, leading to PFOA accumulation in the proximal tubule cells[10].

2.1.3.2. Enterohepatic circulation.

2.2. Model parametrization

The uptake and elimination of PFOA are predicted using in vitro to in vivo extrapolation principles, while its tissue distribution is predicted using experimentally derived distribution coefficients to tissue components, such as albumin and phospholipids. The model introduces a minimal mechanistic kidney compartment to predict the active re-absorption of PFOA in the proximal tubules and a mechanistic transporter-mediated entero-hepatic circulation. Furthermore, given the high affinity of PFOA to albumin, the unbound fraction determined in plasma, is adjusted to the differential albumin concentration between the plasma and tissue or tubular fluid compartments, to better describe an albumin facilitated transport of PFOA.

2.2.1. Derivation of partition coefficients

2.2.2. In Vitro to In Vivo extrapolation

2.2.3. Prediction of the adjusted-fraction unbound

2.3. Model evaluation

2.3.1. Global Sensitivity Analysis Workflow

2.3.1.1. Morris screaning test.

2.3.1.2. eFAST test.

2.3.2. Evaluation against human biomonitoring data

Check this study (Domingo, 2025; Convertino et al., 2018; Hanvoravongchai et al., 2024; Botelho et al., 2025; Lin et al., 2024; Faquih et al., 2024)

3. Results

3.1. Model development

Think if the results of the tissue-plasma partition coefficients are needed here. I think its enough to have them in the supplementary data.

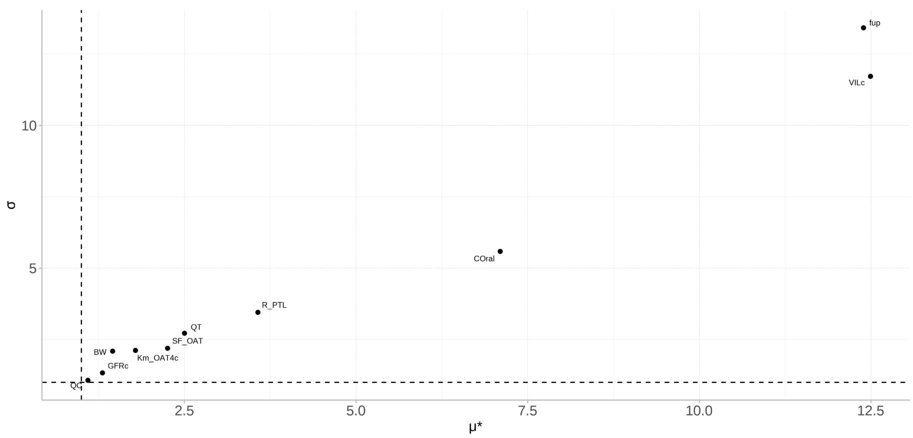
Model figure Show results with & without sub compartments, from most complex to less complex and which processess were included

[10] Not sure if I should further elaborate on this.

3.2. Model evaluation

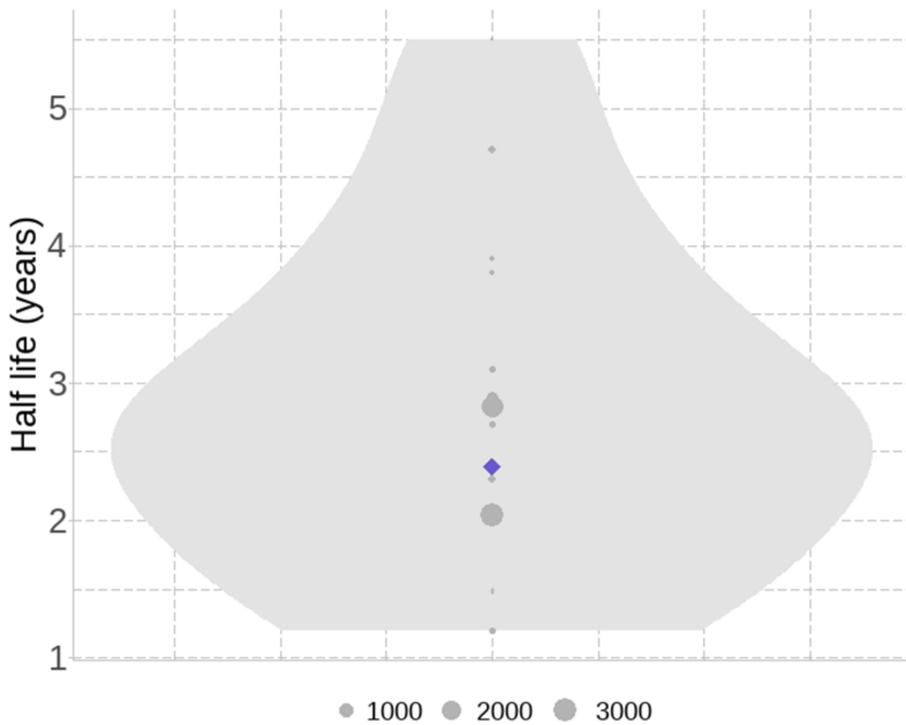
3.2.1. Global Sensitivity Analysis

Finally, the Global Sensitivity Analysis (GSA) workflow recommended by the OECD PBK guideline was applied to both evaluate the validity of the model and inform about the key parameters determining the PBK model predictions.



3.2.2. Evaluation against human biomonitoring data

The predicted Area Under the Curve (AUC) and half-life were compared to the concentration over time curve from a human volunteer study as well as to observed plasma concentrations and half-lives from the latest human biomonitoring data.



4. Discussion

Oral absorption/GIT: differential absorption of PFOA in the GIT depending on it's ionisation status could have been significant if we separately considered passive absorption of the non-ionised form and active absorption of the non-ionised form. But we couldn't do it as mechanistic data on this is not available.

Model limitations : tissue absorption not fully understood, uncertainty of the most sensitive parameters Potential uses?!

The newly developed PFOA PBK model can be extrapolated to other PFASs for which these key parameters are already available. The subsequent derivation of these parameters for other PFASs will facilitate the extrapolation of the present PBK models to them as well and enable their human toxicological risk assessment.

Distribution: Recently, equilibrium tissue distribution coefficients were derived using PFOA tissue concentrations found in healthy human forensic autopsies of different tissues. This approach could be considered more physiologically relevant, however its application would increase the uncertainty of the model some limitations were the sizes of the tissue samples, that might not be representative of the whole tissue concentrations, as well as the variability in blood compositions due to blood coagulation in some of the samples. (Baumert et al., 2023)

Distribution: the need of a fitted parameter in permeability limited models suggest a rapid equilibrium of PFOA concentrations in plasma and cells.

Finally passive processes could be important in determining the half life of PFOA in specific populations, such as patients with chronic kidney disease (CKD), which has actually been linked to PFOA exposure. CKD leads to decreased GFR and changes in urinary pH, both of which alter the passive permeability of PFOA in the kidney (Huang and Isoherranen, 2018) . This is interesting to model, especially for the assessment of the risk of PFOA exposure in that population.

5. Conclusion

6. Funding

7. CRediT authorship contribution statement

8. Declaration of Competing Interest

9. Acknowledgement

10. Appendix A. Supporting information

11. Data availability

References

Abraham, K., Mertens, H., Richter, L., Mielke, H., Schwerdtle, T., Monien, B.H., 2024. Kinetics of 15 per- and polyfluoroalkyl substances (pfas) after single oral application as a mixture – a pilot investigation in a male volunteer. Environment International 193, 109047. URL: http://dx.doi.org/10.1016/j.envint.2024.109047, doi:10.1016/j.envint.2024.109047.

Allendorf, F., Goss, K.U., Ulrich, N., 2020. Estimating the equilibrium distribution of perfluoroalkyl acids and 4 of their alternatives in mammals. Environmental Toxicology and Chemistry 40, 910–920. URL: http://dx.doi.org/10.1002/etc.4954, doi:10.1002/etc.4954.

Baumert, B.O., Fischer, F.C., Nielsen, F., Grandjean, P., Bartell, S., Stratakis, N., Walker, D.I., Valvi, D., Kohli, R., Inge, T., Ryder, J., Jenkins, T., Sisley, S., Xanthakos, S., Rock, S., La Merrill, M.A., Conti, D., McConnell, R., Chatzi, L., 2023. Paired liver:plasma pfas concentration ratios from adolescents in the teen-labs study and derivation of empirical and mass balance models to predict and explain liver pfas accumulation. Environmental Science & Technology 57, 14817–14826. URL: http://dx.doi.org/10.1021/acs.est.3c02765, doi:10.1021/acs.est.3c02765.

Botelho, J.C., Kato, K., Wong, L.Y., Calafat, A.M., 2025. Per- and polyfluoroalkyl substances (pfas) exposure in the u.s. population: Nhanes 1999–march 2020. Environmental Research 270, 120916. URL: http://dx.doi.org/10.1016/j.envres.2025.120916, doi:10.1016/j.envres.2025.120916.

Cheng, W., Ng, C.A., 2017. A permeability-limited physiologically based pharmacokinetic (pbpk) model for perfluorooctanoic acid (pfoa) in male rats. Environmental Science & Technology 51, 9930–9939. URL: http://dx.doi.org/10.1021/acs.est.7b02602, doi:10.1021/acs.est.7b02602.

Convertino, M., Church, T.R., Olsen, G.W., Liu, Y., Doyle, E., Elcombe, C.R., Barnett, A.L., Samuel, L.M., MacPherson, I.R., Evans, T.R.J., 2018. Stochastic pharmacokinetic-pharmacodynamic modeling for assessing the systemic health risk of perfluorooctanoate (pfoa). Toxicological Sciences 163, 293–306. URL: http://dx.doi.org/10.1093/toxsci/kfy035, doi:10.1093/toxsci/kfy035.

Domingo, J.L., 2025. A review of the occurrence and distribution of per- and polyfluoroalkyl substances (pfas) in human organs and fetal tissues. Environmental Research 272, 121181. URL: http://dx.doi.org/10.1016/j.envres.2025.121181, doi:10.1016/j.envres.2025.121181.

Faquih, T.O., Landstra, E.N., van Hylckama Vlieg, A., Aziz, N.A., Li-Gao, R., de Mutsert, R., Rosendaal, F.R., Noordam, R., van Heemst, D., Mook-Kanamori, D.O., van Dijk, K.W., Breteler, M.M.B., 2024. Per- and polyfluoroalkyl substances concentrations are associated with an unfavorable cardiometabolic risk profile: Findings from two population-based cohort studies. Exposure and Health 16, 1251–1262. URL: http://dx.doi.org/10.1007/s12403-023-00622-4, doi:10.1007/s12403-023-00622-4.

Forsthuber, M., Kaiser, A.M., Granitzer, S., Hassl, I., Hengstschläger, M., Stangl, H., Gundacker, C., 2020. Albumin is the major carrier protein for pfos, pfoa, pfhxs, pfna and pfda in human plasma. Environment International 137, 105324. URL: http://dx.doi.org/10.1016/j.envint.2019.105324, doi:10.1016/j.envint.2019.105324.

Gustafsson, ., Wang, B., Gerde, P., Bergman, ., Yeung, L.W.Y., 2022. Bioavailability of inhaled or ingested pfoa adsorbed to house dust. Environmental Science and Pollution Research 29, 78698–78710. URL: http://dx.doi.org/10.1007/s11356-022-20829-3, doi:10.1007/s11356-022-20829-3.

Hanvoravongchai, J., Laochindawat, M., Kimura, Y., Mise, N., Ichihara, S., 2024. Clinical, histological, molecular, and toxicokinetic renal outcomes of per-/polyfluoroalkyl substances (pfas) exposure: Systematic review and meta-analysis. Chemosphere 368, 143745. URL: http://dx.doi.org/10.1016/j.chemosphere.2024.143745, doi:10.1016/j.chemosphere.2024.143745.

Huang, W., Isoherranen, N., 2018. Development of a dynamic physiologically based mechanistic kidney model to predict renal clearance. CPT: Pharmacometrics & Systems Pharmacology 7, 593–602. URL: http://dx.doi.org/10.1002/psp4.12321, doi:10.1002/psp4.12321.

Husøy, T., Caspersen, I.H., Thépaut, E., Knutsen, H., Haug, L.S., Andreassen, M., Gkrillas, A., Lindeman, B., Thomsen, C., Herzke, D., Dirven, H., Wojewodzic, M., 2023. Comparison of aggregated exposure to perfluorooctanoic acid (pfoa) from diet and personal care products with concentrations in blood using a pbpk model – results from the norwegian biomonitoring study in euromix. URL: http://dx.doi.org/10.2139/ssrn.4450658.

Janssen, A.W.F., Duivenvoorde, L.P.M., Beekmann, K., Pinckaers, N., van der Hee, B., Noorlander, A., Leenders, L.L., Louisse, J., van der Zande, M., 2024. Transport of perfluoroalkyl substances across human induced pluripotent stem cell-derived intestinal epithelial cells in comparison with primary human intestinal epithelial cells and caco-2 cells. Archives of Toxicology 98, 3777–3795. URL: http://dx.doi.org/10.1007/s00204-024-03851-x, doi:10.1007/s00204-024-03851-x.

Kimura, O., Fujii, Y., Haraguchi, K., Kato, Y., Ohta, C., Koga, N., Endo, T., 2017. Uptake of perfluorooctanoic acid by caco-2 cells: Involvement of organic anion transporting polypeptides. Toxicology Letters 277, 18–23. URL: http://dx.doi.org/10.1016/j.toxlet.2017.05.012, doi:10.1016/j.toxlet.2017.05.012.

Lin, C.Y., Huey-Jen Hsu, S., Lee, H.L., Wang, C., Sung, F.C., Su, T.C., 2024. Examining a decade-long trend in exposure to per- and polyfluoroalkyl substances and their correlation with lipid profiles: Insights from a prospective cohort study on the young taiwanese population. Chemosphere 364, 143072. URL: http://dx.doi.org/10.1016/j.chemosphere.2024.143072, doi:10.1016/j.chemosphere.2024.143072.

Lin, J., Chin, S.Y., Tan, S.P.F., Koh, H.C., Cheong, E.J.Y., Chan, E.C.Y., Chan, J.C.Y., 2023. Mechanistic middle-out physiologically based toxicokinetic modeling of transporter-dependent disposition of perfluorooctanoic acid in humans. Environmental Science & Technology 57, 6825–6834. URL: http://dx.doi.org/10.1021/acs.est.2c05642, doi:10.1021/acs.est.2c05642.

Peyret, T., Poulin, P., Krishnan, K., 2010. A unified algorithm for predicting partition coefficients for pbpk modeling of drugs and environmental chemicals. Toxicology and Applied Pharmacology 249, 197–207. URL: http://dx.doi.org/10.1016/j.taap.2010.09.010, doi:10.1016/j.taap.2010.09.010.

Pletz, J., Allen, T.J., Madden, J.C., Cronin, M.T., Webb, S.D., 2021. A mechanistic model to study the kinetics and toxicity of salicylic acid in the kidney of four virtual individuals. Computational Toxicology 19, 100172. URL: http://dx.doi.org/10.1016/j.comtox.2021.100172, doi:10.1016/j.comtox.2021.100172.

Rodgers, T., Rowland, M., 2007. Mechanistic approaches to volume of distribution predictions: Understanding the processes. Pharmaceutical Research 24, 918–933. URL: http://dx.doi.org/10.1007/s11095-006-9210-3, doi:10.1007/s11095-006-9210-3.

Ryu, S., Yamaguchi, E., Sadegh Modaresi, S.M., Agudelo, J., Costales, C., West, M.A., Fischer, F., Slitt, A.L., 2024. Evaluation of 14 pfas for permeability and organic anion transporter interactions: Implications for renal clearance in humans. Chemosphere 361, 142390. URL: http://dx.doi.org/10.1016/j.chemosphere.2024.142390, doi:10.1016/j.chemosphere.2024.142390.

Scotcher, D., Jones, C., Rostami-Hodjegan, A., Galetin, A., 2016. Novel minimal physiologically-based model for the prediction of passive tubular reabsorption and renal excretion clearance. European Journal of Pharmaceutical Sciences 94, 59–71. URL: http://dx.doi.org/10.1016/j.ejps.2016.03.018, doi:10.1016/j.ejps.2016.03.018.

Utsey, K., Gastonguay, M.S., Russell, S., Freling, R., Riggs, M.M., Elmokadem, A., 2020. Quantification of the impact of partition coefficient prediction methods on physiologically based pharmacokinetic model output using a standardized tissue composition. Drug Metabolism and Disposition 48, 903–916. URL: http://dx.doi.org/10.1124/dmd.120.090498, doi:10.1124/dmd.120.090498.

Willmann, S., Schmitt, W., Keldenich, J., Lippert, J., Dressman, J.B., 2004. A physiological model for the estimation of the fraction dose absorbed in humans. Journal of Medicinal Chemistry 47, 4022–4031. URL: http://dx.doi.org/10.1021/jm030999b, doi:10.1021/jm030999b.

Yang, C.H., Glover, K.P., Han, X., 2010. Characterization of cellular uptake of perfluorooctanoate via organic anion-transporting polypeptide 1a2, organic anion transporter 4, and urate transporter 1 for their potential roles in mediating human renal reabsorption of perfluorocarboxylates. Toxicological Sciences 117, 294–302. URL: http://dx.doi.org/10.1093/toxsci/kfq219, doi:10.1093/toxsci/kfq219.

Yao, Y., Wang, X., Liu, F., Zhang, W., Artigas, F.J., Gao, Y., 2025. Distributions and partitioning of airborne per- and polyfluoroalkyl substances (pfas) in urban atmosphere of northern new jersey. Science of The Total Environment 970, 179037. URL: http://dx.doi.org/10.1016/j.scitotenv.2025.179037, doi:10.1016/j.scitotenv.2025.179037.